



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

Software Engineering & Agile Methodologies

Sem	VI		
Course Code:	23DC6PCSEA	Total Contact Hours: 26 hours	
L-T-P:	2-0-0	Total Credits:	2

Unit No.	Topics	Hrs.
1	Introduction - Software metrics, Overview of Software Development Projects, Emergence of Software Engineering Software Life Cycle Models - Basic Concepts, Waterfall Model and its Extensions, Rapid Application Development, Agile Development Models: Essential Ideas Behind Agile Models, Agile vs. Other Models, Spiral Model, Comparison of Different Life Cycle Models	5
2	Requirements Analysis and Specification - Requirements Gathering and Analysis, Software Requirements Specification (SRS) Case Study - IEEE SRS Software Design - Overview of the Design Process, Characteristics of Good Software Design: Cohesion and Coupling, Approaches to Software Design Function-Oriented Software Design - Overview of SA/SD Methodology, Structured Analysis: Developing the DFD Model of a System, Structured Design and Detailed Design	5
3	Software Project Management - Project Planning, Metrics for Project Size Estimation, Project Estimation Techniques: Empirical Estimation Techniques, COCOMO (A Heuristic Estimation Technique), Scheduling and Team Structures	5
4	Understanding Agile - What is Agile?, The Agile Manifesto and Principles, Why Agile Works Better than Traditional Models Kanban and Lean - Introduction to Kanban Method, Lean Principles in Agile Jira Fundamentals - Overview of Jira: Project Boards, Enrich Issues, Kanban Boards, Scrum Projects, Quick Search and Basic Search, JQL (Jira Query Language), Filters, Epics, Dashboards	6
5	Understanding XP - The XP Life cycle, The XP Team, XP Concepts, Software Configuration Management - Configuration Management Process, Version Control Systems, Change Management and Control	5

Prescribed Text Book

Sl. No.	Book Title	Authors	Edition	Publisher	Year
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SEE Exam Question paper format

Unit-1	Mandatory	One Question to be asked for 20 Marks
Unit-2	Internal Choice	One Question to be asked for 20 Marks
Unit-3	Internal Choice	One Question to be asked for 20 Marks
Unit-4	Mandatory	One Question to be asked for 20 Marks
Unit-5	Mandatory	One Question to be asked for 20 Marks